

Topic 2

Algebra

**Concept guide · Worked examples · Practice · Investigation and modelling
· Solutions**

Learning focus

Content explanations, original Cambridge-style practice and an investigation task tailored to International Mathematics 0607.

Topic map

2.1	Algebraic notation and expressions
2.2	Expansion and factorisation
2.3	Linear, simultaneous and quadratic equations
2.4	Inequalities and rearranging formulae
2.5	Sequences and direct or inverse variation
2.6	Algebraic fractions and surds

Core foundations and Extended applications are integrated. Practise exact methods, calculator work and the written reasoning required in investigations.

1. Expressions and algebraic rules

Algebra uses letters for unknown or varying values. Like terms have the same variable part; add their coefficients. The distributive law multiplies each term in a bracket. To factorise, identify a common factor or reverse a quadratic expansion. State any excluded values when simplifying an algebraic fraction.

Worked example

$$3(x - 2) - 2(x + 5) = 3x - 6 - 2x - 10 = x - 16.$$

Exam check: Keep the minus sign when expanding a negative bracket.

2. Equations and formulae

An equation is balanced when the same operation is applied to both sides. Eliminate or substitute to solve two equations in two unknowns. Rearranging a formula treats the requested letter as the unknown and every other letter as constant. Check the final values in the original equations.

Worked example

$$2x + y = 10 \text{ and } x - y = 2. \text{ Add equations: } 3x = 12, \text{ so } x = 4, y = 2.$$

Exam check: A correct ordered pair must satisfy both original equations.

3. Quadratic expressions

A quadratic can be solved by factorisation, completing the square, the quadratic formula or numerical graph intersection. If $ab = 0$, either $a = 0$ or $b = 0$. The discriminant $b^2 - 4ac$ signals how many real roots a quadratic has. Use exact forms in a non-calculator question when needed.

Worked example

$$x^2 - 7x + 12 = (x - 3)(x - 4) = 0, \text{ giving } x = 3 \text{ or } x = 4.$$

Exam check: Do not drop the second solution after factorising.

4. Inequalities and intervals

Solve a linear inequality like an equation, reversing its direction if multiplying or dividing by a negative number. Represent solutions with correct open or closed endpoints. A double inequality can constrain both sides. For an integer answer, list only whole values in the interval.

Worked example

$$-2x + 5 < 11 \text{ gives } -2x < 6, \text{ hence } x > -3.$$

Exam check: Division by a negative number reverses the inequality symbol.

5. Sequences and variation

Arithmetic sequences have constant first differences. The n th term is $a + (n - 1)d$. Quadratic sequences have constant second differences. If y is directly proportional to x^2 , write $y = kx^2$; for inverse proportionality use $y = k/x$ or k/x^2 as specified. Find k before predicting.

Worked example

For 7, 11, 15, ... the n th term is $7 + 4(n - 1) = 4n + 3$.

Exam check: Use at least two terms to test a proposed rule.

6. Surds and algebraic fractions

Simplify square roots by extracting perfect-square factors. Rationalise a single-surds denominator by multiplying top and bottom by the radical. Algebraic fractions may be cancelled only after factorisation, and only common factors can cancel. For $x^2 - 9$ over $x - 3$, $x = 3$ remains excluded even after cancellation.

Worked example

$(x^2 - 9)/(x - 3) = (x - 3)(x + 3)/(x - 3) = x + 3$, for $x \neq 3$.

Exam check: Cancelling a term from a sum changes the value; cancel factors only.

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Graphical calculator and Paper 6

Use symbolic algebra for a non-calculator method; use a graph or table to verify roots and the pattern in a sequence.

State the rule or model, display enough working to reproduce your result, test it against the given values and discuss at least one limitation.



Practice paper

Topic 2 · Algebra

Instructions

Original practice, informed by June 2026 components. Section A: short multiple choice for revision without a calculator. Section B: show full written working. Section C: investigation and modelling with reasons and checks.

Candidate name:

Class / date:

Total: 39 marks

Section A · Multiple choice

Choose one answer for each question.

1. Simplify $5a - 2a + 7$. [1]

- A $3a + 7$
- B $7a + 7$
- C $3a - 7$
- D $10a + 7$

2. Expand $-2(x - 3)$. [1]

- A $-2x - 6$
- B $-2x + 6$
- C $2x - 6$
- D $-x + 3$

3. Factorise $x^2 - 16$. [1]

- A $(x - 4)(x + 4)$
- B $(x - 8)^2$
- C $x(x - 16)$
- D $(x - 4)^2$

4. Solve $3x + 2 = 14$. [1]

- A 3
- B 4
- C 5
- D 6

5. Solve $-2x < 8$. [1]

- A $x < -4$
- B $x > -4$
- C $x < 4$
- D $x > 4$

6. Nth term of 3, 8, 13, ...? [1]

- A $3n + 5$
- B $5n - 2$
- C $5n + 2$
- D $8n - 5$

7. If $y \propto x$ and $y = 12$ at $x = 4$, then y at $x = 7$ is [1]

- A 15
- B 18
- C 21
- D 28

8. For which x is $(x^2 - 9)/(x - 3)$ undefined? [1]

- A -3
- B 0
- C 3
- D 9

Section B · Structured questions

9. A rectangle has sides $x + 4$ cm and $x - 2$ cm.

(a) Expand its area as a quadratic. [2]

(b) Find the area when $x = 5$. [2]

(c) Factorise the quadratic to check. [2]

[Total: 6]

10. Solve the equations $2x + 3y = 17$ and $x - y = 1$.

(a) Make x the subject of $x - y = 1$. [1]

(b) Find x and y . [3]

(c) Substitute them into $2x + 3y = 17$.

[1]

[Total: 5]

11. The n th term of a sequence is $5n - 3$.

(a) Find the first three terms.

[2]

(b) Find the value of n when the term is 72.

[2]

(c) State whether 100 belongs to this sequence, with a reason.

[2]

[Total: 6]

12. A quantity y is inversely proportional to x . When $x = 4$, $y = 15$.

(a) Find the constant of proportionality.

[2]

(b) Find y when $x = 10$.

[2]

(c) Find x when $y = 8$.

[2]

[Total: 6]

Section C · Investigation and modelling

Show the rule, test it with values and explain the result in context.

13. A pattern of tiles has counts 4, 9, 16 and 25 for figures $n=1,2,3,4$.

(a) Find the first and second differences. [2]

(b) Suggest an expression $T(n)$ for the tile count. [2]

(c) Use it to predict figure 8. [2]

(d) Explain how you would check the rule against the original figures. [2]

[Total: 8]



Worked solutions

Fold back this section while completing the practice paper.

Marking routine

Method, model, communication and verification matter. A numerical answer should follow from clear working.

Section A · Answers and reasons

1. A — $3a + 7$. $5a - 2a = 3a$, while 7 remains constant.
2. B — $-2x + 6$. Both terms change sign when multiplied by -2.
3. A — $(x - 4)(x + 4)$. Use the difference of squares $(x-4)(x+4)$.
4. B — 4. Subtract 2, then divide 12 by 3.
5. B — $x > -4$. Dividing $-2x < 8$ by -2 reverses the inequality.
6. B — $5n - 2$. The difference is 5, and $5n-2$ starts at 3.
7. C — 21. $k = 12/4 = 3$, so $y = 3(7)$.
8. C — 3. The denominator $x - 3$ is zero when $x = 3$.

Section B · Worked solutions

9. a) $(x + 4)(x - 2) = x^2 + 2x - 8 \text{ cm}^2$ [2]. b) $9 \times 3 = 27 \text{ cm}^2$ [2]. c) $(x + 4)(x - 2)$ [2].

10. a) $x = y + 1$ [1]. b) $2(y + 1) + 3y = 17$, so $y = 3$, $x = 4$ [3]. c) $2(4) + 3(3) = 17$ [1].

11. a) $n=1,2,3$ gives 2,7,12 [2]. b) $5n-3=72$ gives $n=15$ [2]. c) $5n-3=100$ implies $n=20.6$, not a positive integer, so no [2].

12. a) $y=k/x$, $k=15 \times 4=60$ [2]. b) $y=60/10=6$ [2]. c) $8=60/x$ implies $x=7.5$ [2].

Section C · Investigation and modelling solutions

13(a). Differences 5,7,9; second differences 2,2 [2].

13(b). Counts are $(n+1)^2$, so $T(n)=(n+1)^2$ [2].

13(c). $T(8)=81$ [2].

13(d). Substitute $n=1,2,3,4$ and confirm 4,9,16,25; then examine the tile arrangement [2].

Self-check

Explain why your model fits the starting data, show an independent check and name a condition under which your prediction would be unreliable.

Reference: Cambridge IGCSE International Mathematics 0607 syllabus 2025-2027, <https://www.cambridgeinternational.org/Images/662472-2025-2027-syllabus.pdf>. Assessment conventions informed by the June 2026 papers and mark schemes supplied by the user. All practice questions are newly written.